

Cloud Computing-Question Bank

TY 2025-26-Sem-II (PEC-3)

Unit-I: Introduction to Cloud Computing

Low Level of Difficulty (Understanding)

1. Illustrate the concept of multi-tenancy in SaaS and its implications on data security.
2. Elucidate the future trends in cloud computing, particularly in the context of AI and machine learning integration.
3. Describe how virtualization contributes to the scalability of cloud services.
4. Elucidate the significance of Platform as a Service (PaaS) in application development. How does it differ from IaaS?
5. Illustrate the impact of cloud computing on traditional IT infrastructure management.
6. Elaborate the ethical considerations in the use of public cloud services.
7. Describe the advantages and disadvantages of using SaaS in a multi-tenant environment.

Moderate Level of Difficulty (Applying)

1. Which deployment model is applied if an organization wants to protect confidential data & display public data on cloud users? Justify your answer.
2. Applying load balancing enhances the performance and reliability of cloud services? Describe with the examples.
3. Does applying the storage redundancy will ensure the data availability and durability in cloud computing? Illustrate.
4. Discuss the security challenges associated with IaaS and suggest mitigation strategies.
5. How does cloud computing support business continuity and disaster recovery? Provide real-world examples.
6. Discuss the economic implications of adopting a cloud computing model for a large enterprise.
7. How do load balancing algorithm applications differ in public and private cloud environments?
8. Examine the challenges or potential drawbacks that organizations should take into account while implementing infrastructure as service.,
9. Categorize the public, private, and hybrid clouds. and also provide an illustrative example for each type of cloud deployment?
10. Analyze the three primary cloud service models (IaaS, PaaS, SaaS) and their respective benefits

11. Illustrate an example of real-world scenarios where load balancing, system redundancy or storage redundancy helped mitigate performance issues or service outages in cloud computing?
12. Distinguish clouds from traditional IT infrastructure with an example.
13. How do horizontal and vertical scalability in cloud computing differ from each other?
14. Determine the challenges or potential drawbacks that organizations should take into account while implementing strategies in their cloud infrastructure.
15. Distinguish the various cloud service models and their respective benefits.

Higher Order Thinking Skills (HOTS - Analyzing, Evaluating, Creating)

1. Compare and contrast public, private and hybrid cloud deployment models in terms of security, scalability and cost-effectiveness.
2. Analyze the impact of service-oriented architecture (SOA) on cloud computing service models.
3. Analyze the potential risks associated with data migration to the cloud. How can these risks be mitigated?
4. Compare the scalability features of SaaS, PaaS and IaaS. Which service model is most suited for a rapidly growing startup?
5. Evaluate the role of system redundancy in disaster recovery planning for cloud-based services.
6. How does cloud computing enable global collaboration? Provide examples of industries that benefit the most.
7. **In a case study, a company wants to build a SaaS application with multi-tenant architecture. What are the key considerations in designing the service to ensure data isolation and scalability?**
8. **Analyze how the redundancy in storage and systems help ensure high availability in cloud environments, particularly in IaaS models?**
9. Examine the Service-Oriented Architecture and how does it interact with cloud computing models, particularly in PaaS environments?

Unit-II: Web Services

Low Level of Difficulty (Understanding)

1. Discuss how web services enable cloud computing platforms like AWS to function effectively.
2. Describe the architecture of Amazon Web Services and how its components interact.
3. Illustrate the working principle of Elastic Compute Cloud (EC2) in managing virtual servers.
4. Interpret the role of AWS storage services in handling structured and unstructured data.
5. Elaborate Platform as a Service (PaaS) simplifies application development and deployment.
6. Compare SaaS and PaaS in terms of user control and service responsibility.
7. Illustrate how AWS ensures scalability and elasticity through its services.
8. Describe how AWS regions and availability zones contribute to system reliability, and apply these concepts to design a highly available application deployment.

Moderate Level of Difficulty (Applying)

1. Apply AWS services to set up a basic web application infrastructure.
2. Demonstrate the steps to launch and configure an EC2 instance for a real-world application.
3. Apply appropriate AWS storage services for a media streaming application and justify your selection.
4. Use EC2 features to scale an application during high traffic conditions.
5. Apply AWS concepts to design a backup solution using storage services.
6. Illustrate how PaaS can be used to deploy a web application without managing infrastructure.
7. Apply SaaS and PaaS models in different business scenarios and explain their suitability.
8. Use AWS storage services to manage large-scale data efficiently.
9. Demonstrate how load balancing can be implemented using AWS services.
10. Apply AWS security features while configuring EC2 instances.
11. Use AWS services to ensure high availability in a distributed application.
12. Apply EC2 pricing models to select a cost-effective solution for a startup.
13. Demonstrate how to store and retrieve data using AWS S3.
14. Apply AWS concepts to implement disaster recovery for a small enterprise.
15. Use PaaS solutions to reduce development time in a project.
16. Apply AWS monitoring tools to track system performance.
17. Demonstrate how AWS services can be used to deploy scalable APIs.
18. Apply cloud service models (SaaS, PaaS) in an educational platform scenario.
19. Use AWS storage options to optimize performance and cost for a database-driven application.
20. Apply AWS architecture principles to design a fault-tolerant system.

Higher Order Thinking Skills (HOTS - Analyzing, Evaluating, Creating)

1. Analyze the effectiveness of AWS EC2 compared to traditional server infrastructure in dynamic workloads.
2. Evaluate different AWS storage services and recommend the best solution for a big data analytics system.
3. Analyze the trade-offs between SaaS and PaaS for enterprise application deployment.
4. A national bank is planning to migrate its core banking system to the cloud. The system must handle millions of transactions per day, ensure zero downtime, and maintain data integrity even during server or network failures. Design a scalable and fault-tolerant architecture using AWS services for this scenario and justify your design choices.
5. A software development company is adopting DevOps practices to accelerate product delivery. The team wants to automate build, testing, and deployment processes while minimizing infrastructure management overhead. Evaluate how Platform as a Service (PaaS) can support modern DevOps practices and continuous deployment pipelines in this scenario.
6. Create a multi-tier cloud architecture using AWS components and justify service selection.

Unit-3

Database Services in Cloud Computing

Low Level of Difficulty (Understanding)

1. What is a cloud database? Compare it with traditional database systems in terms of architecture, cost, scalability, and maintenance. Also discuss the need for cloud databases in modern applications.
2. Discuss the key characteristics of cloud databases such as scalability, availability, performance, and security. How do these characteristics make cloud databases suitable for large-scale applications?
3. What is Database-as-a-Service (DBaaS)? Compare DBaaS with database deployment on cloud virtual machines in terms of control, cost, scalability, and maintenance. Justify which approach is more suitable for modern enterprises.
4. Discuss the architecture of cloud databases. How does distributed architecture contribute to high availability and fault tolerance? Support your answer with suitable examples.

5. Classify different cloud database models such as relational, key–value, document-based, graph-based, and time-series databases. Compare them based on structure, scalability, and use cases with examples.
6. Relational databases and NoSQL databases serve different application needs. Compare these two models in terms of schema, consistency, scalability, and performance. Justify their suitability with real-world examples.
7. Discuss the concept of multi-zone replication in cloud databases. How does it ensure high availability and fault tolerance? Explain the role of primary and replica instances in this architecture.
8. Explain the concept of partitioning (sharding) and replication in cloud databases. How do these techniques improve scalability, availability, and performance? Support your answer with suitable examples.
9. Discuss the working of in-memory cloud databases. Evaluate the role of caching strategies such as write-through and write-back in improving system performance.
10. A banking system requires high consistency, security, and reliability. Select a suitable cloud database model and justify your choice based on ACID properties, data integrity, and transaction management.
11. An e-commerce application needs to handle millions of users and real-time data access. Design a suitable cloud database solution by selecting appropriate database models and justify your design choices.
12. Discuss the advantages and limitations of DBaaS. How does it impact cost, performance, and operational efficiency for organizations handling large-scale data?
13. Analyze the role of cloud databases in ensuring business continuity and disaster recovery. How do replication and distributed storage contribute to this?
14. Discuss different NoSQL database models such as key–value, document-based, and graph databases. Compare their strengths and limitations with suitable real-world applications.
15. A social media platform handles highly dynamic and unstructured data. Design a cloud database architecture suitable for this application and justify the selection of database types and techniques used.

Moderate Level of Difficulty (Applying)

1. Identify the major database services available in AWS and discuss their suitability for different application scenarios.

2. Examine how Amazon RDS simplifies relational database management in cloud environments compared to traditional approaches.
3. Evaluate the features of Amazon DynamoDB that support high-performance and low-latency applications.
4. Analyze how DynamoDB achieves scalability through distributed architecture and data partitioning.
5. Discuss the advantages of using managed database services in AWS for modern application development.
6. Assess the role of Multi-AZ deployment in AWS for ensuring high availability and fault tolerance in database systems.
7. Classify AWS database services into relational and non-relational categories and relate them to appropriate use cases.
8. Compare Amazon RDS and DynamoDB with respect to scalability, consistency, and performance requirements.
9. Examine the process involved in deploying a database using AWS cloud services and the key considerations involved.
10. Analyze the security features provided by AWS for protecting cloud-hosted databases.
11. Evaluate the trade-offs between using database services on AWS EC2 instances and fully managed services like RDS.
12. Discuss how caching services such as Amazon ElastiCache contribute to performance optimization in cloud applications.
13. Assess the role of AWS database services in handling large-scale data processing and analytics workloads.
14. Analyze the factors influencing the selection of an appropriate database service in AWS for a given application.

Higher Order Thinking Skills (HOTS - Analyzing, Evaluating, Creating)

1. A global e-commerce platform faces heavy traffic during sale events, causing performance degradation and downtime. Design a suitable cloud database architecture to handle such situations. Justify your design with respect to scalability, availability, and performance.
2. A banking system requires accurate transaction processing with high security and no data loss. Propose a cloud database solution for this system and justify how consistency, integrity, and reliability are maintained.

3. A social media application stores large volumes of user-generated content such as posts, comments, and media files. Suggest an appropriate cloud database approach and justify the choice based on data type, scalability, and flexibility.
4. A real-time online gaming application requires fast data access and low latency for millions of users. Design a cloud database solution and justify how performance is optimized using suitable techniques.
5. An IoT-based system continuously collects time-stamped sensor data from various devices. Propose a cloud database design suitable for this application and justify your choice based on data handling and analysis requirements.
6. A financial system needs to detect fraud by analyzing relationships between users and transactions. Suggest a suitable cloud database model and justify how it supports efficient relationship analysis.
7. Compare different cloud database models such as relational, key-value, document-based, and graph databases, and analyze their suitability under public, private, and hybrid cloud deployment models.
8. Analyze how the choice of cloud deployment model influences the selection of an appropriate database model for different application scenarios. Support your answer with examples.
9. A large enterprise plans to deploy its applications using a hybrid cloud model. Design a suitable database strategy by selecting appropriate cloud database models and justify your choices.
10. Evaluate the advantages and limitations of deploying relational and NoSQL databases in public and private cloud environments.

Unit 4

Security in Cloud Computing:

1. Understanding Level (Basic Conceptual Questions)

1. Explain the concept of cloud computing security.
2. What is meant by risk in cloud computing?
3. Define risk management in cloud environments.
4. What is enterprise-wide risk management (ERM)?
5. List and explain the types of risks in cloud computing.
6. What are the key security issues in cloud computing?

7. Explain the meaning of data security in cloud computing.
8. What is a cloud digital persona?
9. Define the CIA triad (Confidentiality, Integrity, Availability).
10. What are high availability and scalability services in cloud computing?

2. Moderate Level (Application-Based Questions)

1. How would you apply risk management techniques to secure a cloud-based application?
2. Illustrate how an organization can implement enterprise-wide risk management in cloud infrastructure.
3. Apply the CIA triad principles to secure a cloud storage system.
4. How can authentication and authorization mechanisms be implemented in cloud security?
5. Demonstrate how data encryption can be used to enhance cloud data security.
6. Apply secure cloud software requirements while designing a cloud-based system.
7. How would you handle data breaches in a cloud environment?
8. Explain how secure cloud software testing can be performed in real-time applications.
9. Apply strategies to improve high availability in cloud services.
10. How can scalability be achieved while maintaining security in cloud computing?

3. Difficult Level (Analyze & Create Questions)

1. Analyze different types of risks in cloud computing and propose mitigation strategies.
2. Compare traditional security vs cloud security challenges and evaluate their impact.

3. Analyze the security challenges in authorization mechanisms in cloud computing.
4. Evaluate the advantages and disadvantages of cloud data security models.
5. Design a secure cloud architecture ensuring confidentiality, integrity, and availability.
6. Analyze the role of digital persona in cloud security and its potential risks.
7. Create a risk management framework for a multi-cloud environment.
8. Evaluate different cloud security services and recommend the best combination for an enterprise.
9. Design a secure cloud software testing strategy for a large-scale application.
10. Analyze how high availability and scalability impact cloud security and propose an optimized model.